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Total No. of Questions : 4]

[Total No. of Printed Pages : 12

12thARF(S/Z)JD-2026

1421-B

CHEMISTRY

Time : 3 Hours]

[Maximum Marks : 70

SECTION-A

(OBJECTIVE TYPE QUESTIONS)

(10×1=10)

1. Choose the correct answer and write in your answer book.
- (i) Ideal solution is formed when its components:
- (a) Have zero heat of mixing only
 - (b) Have zero volume change on mixing only
 - (c) Have zero heat of mixing and zero volume change
 - (d) None of the above

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Turn Over

- (ii) Which of the following will form an ideal solution?
- (a) $\text{C}_2\text{H}_5\text{OH}$ and Water
 - (b) HNO_3 and Water
 - (c) CHCl_3 and CH_3COCH_3
 - (d) C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$
- (iii) The electricity required in coulomb for the oxidation of 1 mol of FeO to Fe_2O_3
- (a) 96500 C
 - (b) 2×96500 C
 - (c) 3×96500 C
 - (d) 4×96500 C

(iv) Europium is a:

- (a) S-block element
- (b) P-block element
- (c) d-block element
- (d) f-block element

(v) Which is the first man-made element?

- | | |
|--------|--------|
| (a) Sc | (b) Tc |
| (c) Os | (d) Pt |

(vi) The position of -F in the compound $\text{CH}_3\text{CH}=\text{CHC}(\text{F})(\text{CH}_3)_2$ can be classified as :

- | | |
|-----------|------------|
| (a) Allyl | (b) Aryl |
| (c) Vinyl | (d) Benzyl |

(vii) Ethers are isomeric with:

(a) Aldehydes

(b) Carboxylic acids

(c) Alcohols

(d) Ketones

(viii) In the carbonyl group, the carbon atom undergoes:

(a) SP hybridisation

(b) SP^2 hybridisation

(c) SP^3 hybridisation

(d) SP^3d hybridisation

(ix) The carbyl-amine reaction is given by compounds having:

(a) $-NH_2$ group

(b) $-NH$ group

(c) $-NO_2$ group

(d) $-N$ group

(x) Vitamin C is

(a) Ascorbic acid

(b) Retinol

(c) Lactic acid

(d) Citric acid

SECTION-B

(VERY SHORT ANSWER TYPE QUESTIONS)

(9×2=18)

2. (i) Calculate mole fraction of benzene in solution containing 30% by mass in carbon tetrachloride.
- (ii) The $t_{1/2}$ of a reaction is doubled as the initial concentration of the reactant is doubled. What is the order of the reaction?
- (iii) Indicate the type of isomerism exhibited by the following complex $[Co(en)_3]Cl_3$ and draw the structures for these isomers.
- (iv) Give the uses and environmental effects of DDT.
- (v) What is Arrhenius equation? What is its importance?

(vi) Define the following with an example.

(a) Coordination compound

(b) Ligand

(vii) Describe Reimer-Tiemann reaction.

(viii) Write short note on acidity of carboxylic acids.

(ix) What is Diazotisation? Write the chemical reaction involved in it.

SECTION-C

(SHORT ANSWER TYPE QUESTIONS)

(9×3=27)

3. (i) What is depression in freezing point of a solution?
Show that depression in freezing point is a colligative property.

- (ji) What is electrolysis? State and explain Faraday's first law of electrolysis.
- (jii) What is order of a reaction? Derive expression for the rate constant for the first order reaction.
- (iy) Give important uses and applications of coordination compounds.
- (v) What happens when
- (a) Ethyl chloride is treated with aqueous KOH
 - (b) Chlorobenzene is treated with methyl chloride in the presence of Anhydrous AlCl_3
 - (c) Bromobenzene is treated with methyl bromide and sodium in the presence of dry ether.

(vi) Write the mechanism of acid dehydration of ethanol to yield ethene.

(vii) Describe a method (Hinsberg test) for the identification of Primary, Secondary and Tertiary amines. Also write chemical equations of the reactions involved.

(viii) (a) What do you understand by the term glycosidic linkage? <https://www.jkboseonline.com>

(b) Which sugar is called invert sugar? Why is it called so?

(ix) Write the important structural and functional differences between DNA and RNA.

SECTION-D
(LONG ANSWER TYPE QUESTIONS)

(3×5=15)

- 4/ (i) What is an electrochemical cell? Discuss the construction and working of Galvanic cell. How is Galvanic cell represented?

OR

Define Corrosion. Explain electrochemical theory of rusting. Name the important factors affecting corrosion.

- (ii) Discuss the following properties of transition (3d) elements.

(a) Magnetic Properties

(b) Oxidation States

OR

Indicate the steps in the preparation of KMnO_4 from Pyrolusite ore. Write the ionic equations for the reaction of KMnO_4 in acidic medium with

(a) Iron (II) solution

(b) Oxalate ion

(iii) How will you prepare aldehydes from

(a) Ozonolysis of alkenes

(b) Acyl chloride (Rosenmund reduction)

(c) Nitriles (Stephen reaction)

(d) Toluene (Etard reaction)

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OR

Write short notes (along with the reaction) on:

- (a) Cross aldol condensation
- (b) Cannizaro reaction
- (c) Hell-Volhard-Zelinsky reaction.
